Practice workbook · Name the parts of a nucleotide and the DNA backbone

Labelling phosphate, deoxyribose and base on a nucleotide diagram, naming the repeating unit, and identifying backbone or rung components.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-NAME-NUCLEOTIDE-PARTS

1. Question 1 · part c · 3 marks

June 2022 Paper 1

Worked-example retry · Attempt part c only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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1. Question 1 · part c · 3 marks

June 2022 Paper 1

Worked-example retry · Attempt part c only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

(1)

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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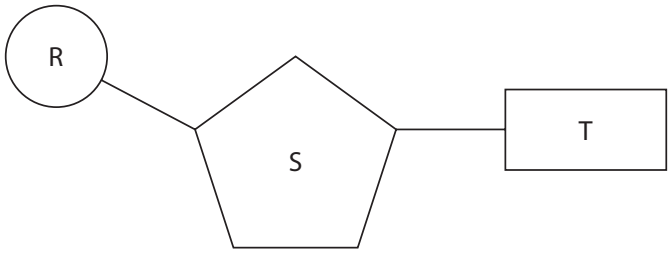
1. Question 1 · part c · 3 marks

June 2022 Paper 1

Worked-example retry · Attempt part c only.

(c) DNA is formed from molecules called nucleotides.

The diagram shows a single nucleotide.



Name the parts of the nucleotide labelled R, S and T.

(3)

R .....

S .....

T .....

(Total for Question 1 = 11 marks)

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Working and answer

2. Question 8 · part a.v · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.v only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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2. Question 8 · part a.v · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.v only.

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(v) Describe the structure of a nucleotide found in a DNA molecule.

(3)

(b) There are 20 different amino acids that can be joined to form a polypeptide chain.

The table gives some base sequences that code for five of these amino acids.

Code	Name of amino acid
AAT or AAC	asparagine
TAT or TAC	tyrosine
TTT or TTC	phenylalanine
CGT or CGG	arginine
CAT or CAC	histidine

Diagram 1 shows part of the gene that codes for a CFTR protein that functions normally.

Diagram 2 shows the same part of the gene with a mutation that causes cystic fibrosis.

AATATCATCTTTGGTGTTTCCTATGAT

Diagram 1

AATATCATCGGTGTTTCCTATGAT

Diagram 2

Describe the changes that occur in the gene and the protein that cause cystic fibrosis.

(3)

(Total for Question 8 = 14 marks)

2. Question 8 · part a.v · 3 marks · Working and answer

Working and answer

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3. Question 1 · part a.i · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.i only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
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- ☒ **C** proteins
- ☒ **D** sugars

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3. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 1 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows part of a molecule of DNA.



(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

(1)

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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4. Question 1 · part a.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part a.ii only.

(iii) Describe how the two backbones of the DNA molecule are joined together.

(2)

(iv) State the name of the part of the cell where most of the DNA is found.

(1)

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

(2)

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

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- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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4. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

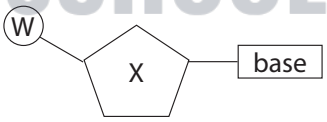
(ii) Give a structure found in bacterial cells.

(1)

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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5. Question 1 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

(ii) Describe how bases hold the two chains of nucleotides together.

(3)

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.....

(c) The gene for the hormone insulin contains 153 bases.

Determine the number of amino acids in insulin.

(2)

number of amino acids = ...

(Total for Question 1 = 11 marks)

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5. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

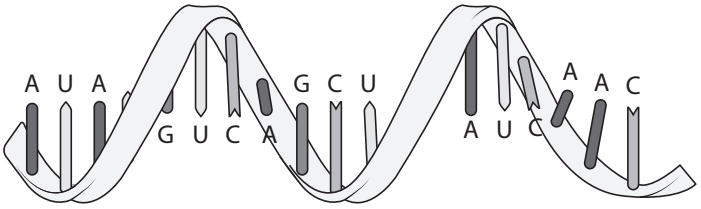
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6. Question 2 · part a.iii · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.iii only.

2 (a) The diagram shows a strand of RNA found in a body cell.



(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

(2)

1 .....

2 .....

(ii) State two differences between RNA and DNA.

(2)

1 .....

.....

2 .....

.....

(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

(1)

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6. Question 2 · part a.iii · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.iii only.

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(b) Before a cell divides by mitosis, the DNA inside the cell is replicated.

When DNA is replicated, mutations can happen.

Explain how mutations can affect the final product made.

(3)

(Total for Question 2 = 8 marks)

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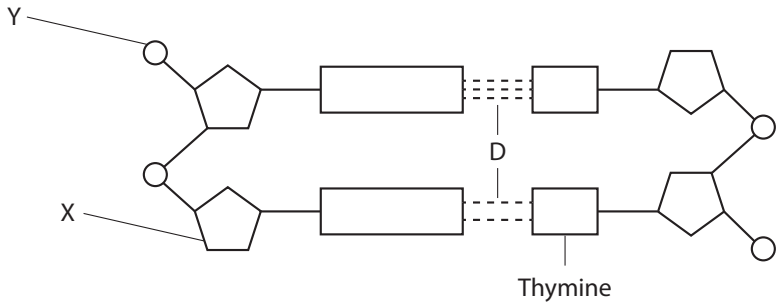
Working and answer

7. Question 2 · part a.i · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

2 The diagram shows part of a molecule of nucleic acid.



(a) (i) Name the parts labelled X and Y.

(2)

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

(1)

.....

(iii) Use information from the diagram to justify your answer to part (ii).

(3)

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(iv) State where this nucleic acid is found in a human cell.

(1)

.....

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7. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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